

| Curriculum |                                 |                           |                                                                     |                                                       |              |
|------------|---------------------------------|---------------------------|---------------------------------------------------------------------|-------------------------------------------------------|--------------|
| Modules    | Topic                           | Sub topic                 | Detail                                                              | Outcome                                               | Duration Min |
| 1          | Introduction to Algorithms      | Algorithm Basics          | Definition, characteristics, importance                             | Understand fundamental concepts of algorithms         | 45           |
|            | Algorithm Efficiency            | Parameters for efficiency | Time and space complexity                                           | Ability to evaluate algorithm efficiency              | 45           |
|            | Complexity Analysis             | Asymptotic notations      | Big O, Theta, Omega notations                                       | Proficiency in using asymptotic notations             | 45           |
|            | Calculating Complexity          | Common rules for Big O    | Loops, nested loops, recursion                                      | Skill in calculating time complexity                  | 45           |
|            | Complexity Functions            | Types of Big O functions  | $O(1)$ , $O(\log n)$ , $O(n)$ , $O(n \log n)$ , $O(n^2)$ , $O(2^n)$ | Familiarity with common complexity functions          | 45           |
|            | Time vs Space Complexity        | Trade-offs                | When to prioritize time or space                                    | Ability to make informed algorithm design choices     | 45           |
| 2          | Searching Algorithms            | Linear Search             | Implementation and analysis                                         | Understand and implement linear search                | 30           |
|            |                                 | Binary Search             | Implementation and analysis                                         | Understand and implement binary search                | 45           |
|            | Sorting Algorithms              | Bubble Sort               | Implementation and analysis                                         | Understand and implement bubble sort                  | 30           |
|            |                                 | Selection Sort            | Implementation and analysis                                         | Understand and implement selection sort               | 30           |
|            |                                 | Insertion Sort            | Implementation and analysis                                         | Understand and implement insertion sort               | 30           |
|            |                                 | Merge Sort                | Implementation and analysis                                         | Understand and implement merge sort                   | 45           |
|            |                                 | Quick Sort                | Implementation and analysis                                         | Understand and implement quick sort                   | 45           |
|            | Analysis of Sorting Algorithms  | Comparison of algorithms  | Time and space complexities, best/average/worst cases               | Ability to choose appropriate sorting algorithm       | 45           |
| 3          | Introduction to Data Structures | Overview                  | Types and importance of data structures                             | Understand the role of data structures in programming | 45           |
|            | Arrays                          | Basic operations          | Insertion, deletion, traversal                                      | Proficiency in using arrays                           | 30           |
|            | Linked Lists                    | Types and representations | Singly, doubly, circular linked lists                               | Understand and implement linked lists                 | 45           |
|            |                                 | Operations                | Insertion, deletion, traversal                                      | Proficiency in linked list operations                 | 45           |
|            | Stacks                          | Basic concepts            | LIFO principle, operations                                          | Understand stack data structure                       | 30           |
|            |                                 | Applications              | Expression evaluation, conversion                                   | Ability to use stacks in problem-solving              | 45           |
| 4          | Queues                          | Basic concepts            | FIFO principle, operations                                          | Understand queue data structure                       | 30           |
|            |                                 | Types                     | Circular queue, dequeue                                             | Familiarity with queue variations                     | 30           |
|            | Trees                           | Basic concepts            | Terminology, types of trees                                         | Understand tree data structure                        | 45           |
|            | Binary Trees                    | Types and traversals      | In-order, pre-order, post-order                                     | Proficiency in binary tree operations                 | 45           |
|            | Binary Search Trees             | Operations                | Insertion, deletion, searching                                      | Ability to implement and use BST                      | 45           |
|            | Balanced Trees                  | AVL Trees                 | Balance factor, rotations                                           | Understand self-balancing trees                       | 45           |
|            |                                 | B and B+ Trees            | Multi-way search trees                                              | Familiarity with database-oriented trees              | 45           |
| 5          | Graphs                          | Basic concepts            | Directed vs undirected, representations                             | Understand graph data structure                       | 45           |
|            | Graph Traversal                 | BFS and DFS               | Implementation and analysis                                         | Proficiency in graph traversal algorithms             | 45           |
|            | Graph Applications              | Shortest path, MST        | Dijkstra's, Prim's algorithms                                       | Ability to solve graph-based problems                 | 45           |
| 6          | File Organization               | Sequential files          | Basics and operations                                               | Understand sequential file organization               | 30           |
|            |                                 | Indexed sequential        | ISAM structure                                                      | Familiarity with indexed sequential access            | 30           |
|            |                                 | Direct access             | Hashing techniques                                                  | Understand direct access file organization            | 30           |
|            | Advanced Topics                 | Dynamic Programming       | Concept and examples                                                | Introduction to dynamic programming                   | 45           |
|            |                                 | Greedy Algorithms         | Concept and examples                                                | Introduction to greedy algorithms                     | 45           |
|            |                                 | Algorithm design          | Applying DSA to solve problems                                      | Ability to design efficient algorithms                | 45           |